

Tiêu đề Title¹

Sample book subtitle²

FIRST-NAME LAST-NAME³

6th August 2026

¹This is a footnote.
²This is yet another footnote.
³www.example.com

Dedicated to Calvin and Hobbes.

Contents

1	Project Introduct	3
1.1	Overview	3
1.1.1	Project Information	3
1.1.2	Project Information	3
1.1.3	Project Team	3
1.2	Project Scope & Limitations	4
1.2.1	Major Features	4
1.2.2	Limitations & Exclusions	6
2	Project Management Plan	7
2.1	Overview	7
2.1.1	Scope and Estimation	7
2.1.2	Project Objectives	10
2.1.3	Project Risks	11
2.2	Management Approach	13
2.2.1	Project Process	13
2.2.2	Quality Management	14
2.2.3	Training Plan	17
2.3	Project Deliverables	17
2.4	Responsibility Assignments	19
2.5	Project Communications	19
2.6	Configuration Management	20
2.6.1	Document Management	20
2.6.2	Source Code Management	22
2.6.3	Tools and Infrastructure	22
2.6.4	Baseline and Change Control	22
3	Software Requirement Specification	24
3.1	Product Overview	24
3.2	Context Diagram	24
3.3	User Requirements	25
3.4	Actors	26
3.5	Section heading	29

3.5.1	Lorem ipsum dolor sit amet, consectetur adipiscing elit.	29
3.5.2	Lorem ipsum dolor sit amet, consectetur adipiscing. .	30
3.5.3	Lorem ipsum dolor sit amet	30
3.5.4	Lorem ipsum dolor sit amet, auctor et pulvinar non .	30
3.6	Another section heading	31
3.6.1	Lorem ipsum dolor sit amet, consectetur adipiscing elit.	32
3.6.2	Lorem ipsum dolor sit amet, consectetur adipiscing . .	32

List of Figures

3.1	Context Diagram of the AI-powered Film Photography Contest Management Platform	25
3.2		28

List of Tables

2.2	Preliminary Defect Distribution Targets	11
2.3	Project Risk Register	11
2.4	Training Plan	16
2.6	Responsibility Assignment Matrix	20
2.7	Project Communication Plan	21
2.8	Tools and Infrastructure	23
3.1	Sample table	31

Preface

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna.

Un-numbered sample section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis.

Another sample section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam. Etiam ullamcorper, nunc a accumsan adipiscing, turpis odio bibendum erat, id convallis magna eros nec metus.

Structure of book

Each unit will focus on <SOMETHING>.

About the companion website

The website¹ for this file contains:

- A link to (freely downloadable) latest version of this document.
- Link to download LaTeX source for this document.
- Miscellaneous material (e.g. suggested readings etc).

Acknowledgements

- A special word of thanks goes to Professor Don Knuth² (for T_EX) and Leslie Lamport³ (for L^AT_EX).
- I'll also like to thank Gummi⁴ developers and LaTeXila⁵ development team for their awesome L^AT_EX editors.
- I'm deeply indebted my parents, colleagues and friends for their support and encouragement.

Amber Jain

<http://amberj.devio.us/>

¹<https://github.com/amberj/latex-book-template>

²<http://www-cs-faculty.stanford.edu/~uno/>

³<http://www.lamport.org/>

⁴<http://gummi.midnightcoding.org/>

⁵<http://projects.gnome.org/latexila/>

1

Project Introduce

1.1 Overview

1.1.1 Project Information

1.1.2 Project Information

- **Project name (English):** AI-powered Film Photography Contest Management Platform
- **Project name (English):** AI-powered Film Photography Contest Management Platform
- **Project name (Vietnamese):** Nền tảng tổ chức và quản lý cuộc thi nhiếp ảnh phim tích hợp trí tuệ nhân tạo
- **Group name:** *Nhóm CNPM*
- **Group name:** *AI Film Photography Team*
- **Software Type:** Web application & Mobile application

1.1.3 Project Team

Full Name	Role	Email	Student ID
<i>Nguyễn Văn Chiến</i>	Supervisor	chiennv@ut.edu.vn	-
<i>Nguyễn Mai Tường Vy</i>	Leader	Vynmt753892@ut.edu.vn	075307023892
<i>Phan Huỳnh Trúc Loan</i>	Member	loanpht750003@ut.edu.vn	075307010003
<i>Lê Nguyễn Quyền Linh</i>	Member	linhlnq823632@ut.edu.vn	082207003632
<i>Nguyễn Tấn Phúc</i>	Member	nguyentanphuc04092007@gmail.com	075207022696
<i>Nguyễn Tấn Tài</i>	Member	taint797744@ut.edu.vn	079207007744

@@ -59,39 +61,39 @@

1.2 Project Scope & Limitations

1.2.1 Major Features

Major Features for Admin

- **FE-01:** Login/Logout
- **FE-02:** Manage user accounts and access permissions
- **FE-03:** Configure system settings and master data
- **FE-04:** Monitor overall platform activities and contest configurations
- **FE-ADM-01:** Login/Logout
- **FE-ADM-02:** Manage user accounts and access permissions
- **FE-ADM-03:** Configure system settings and master data
- **FE-ADM-04:** Monitor overall platform activities and contest configurations

Major Features for Contest Organizer

- **FE-01:** Login/Logout
- **FE-02:** Create and configure photography contests (themes, categories, timelines, rules, award structures)
- **FE-03:** Define custom judging criteria specifically for film photography
- **FE-04:** Manage participant registrations and verify submissions
- **FE-05:** Assign judges to categories or specific evaluation rounds
- **FE-06:** Utilize AI tools for duplicate detection, AI-generated image detection, and automatic categorization
- **FE-07:** Publish official contest results, produce statistical analytics, and manage digital exhibitions
- **FE-ORG-01:** Login/Logout
- **FE-ORG-02:** Create and configure photography contests (themes, categories, timelines, rules, award structures)
- **FE-ORG-03:** Define custom judging criteria specifically for film photography

- **FE-ORG-04:** Manage participant registrations and verify submissions
- **FE-ORG-05:** Assign judges to categories or specific evaluation rounds
- **FE-ORG-06:** Utilize AI tools for duplicate detection, AI-generated image detection, and automatic categorization
- **FE-ORG-07:** Publish official contest results, produce statistical analytics, and manage digital exhibitions

Major Features for Judge

- **FE-01:** Login/Logout
- **FE-02:** Review assigned submissions and view technical film metadata
- **FE-03:** Evaluate photographs using configurable criteria (composition, film grain, tonal range, scan quality)
- **FE-04:** Provide detailed comments, scores, and participate in multiple judging rounds
- **FE-05:** Confirm final evaluation results
- **FE-JDG-01:** Login/Logout
- **FE-JDG-02:** Review assigned submissions and view technical film metadata
- **FE-JDG-03:** Evaluate photographs using configurable criteria (composition, film grain, tonal range, scan quality)
- **FE-JDG-04:** Provide detailed comments, scores, and participate in multiple judging rounds
- **FE-JDG-05:** Confirm final evaluation results

Major Features for Participant

- **FE-01:** Account registration and personal profile management
- **FE-02:** Browse and register for active film photography contests
- **FE-03:** Submit film photographs along with detailed technical metadata
- **FE-04:** Manage film roll history, individual frames, and personal contest portfolio

- **FE-05:** Track submission verification status and receive judging feedback and results
- **FE-PAR-01:** Account registration and personal profile management
- **FE-PAR-02:** Browse and register for active film photography contests
- **FE-PAR-03:** Submit film photographs along with detailed technical metadata
- **FE-PAR-04:** Manage film roll history, individual frames, and personal contest portfolio
- **FE-PAR-05:** Track submission verification status and receive judging feedback and results

1.2.2 Limitations & Exclusions

2

Project Management Plan

2.1 Overview

2.1.1 Scope and Estimation

This chapter presents the management plan for the *AI-powered Film Photography Contest Management Platform*. The project aims to develop a web-based platform that supports the complete operation of film photography contests, including contest configuration, participant registration, film photograph submission, metadata management, submission verification, judging, result publication, and digital archiving.

The system will serve four main user roles: Administrator, Contest Organizer, Judge, and Participant. The project also includes AI-assisted functions for duplicate image detection, AI-generated image detection, automatic image categorization, and contest analytics. The mobile application is excluded from the current project scope.

The following Work Breakdown Structure (WBS) provides a preliminary estimation of project effort. The estimates may be adjusted after the project schedule, frontend technology, and AI models are officially confirmed.

#	WBS Item	Complexity	Estimated Effort (man-days)
1	Initiating		18
1.1	Define project scope, objectives, stakeholders, and limitations	Medium	8
1.2	Collect and consolidate business and user requirements	Medium	10
2	Planning		16
2.1	Conduct project kick-off meeting	Simple	4

#	WBS Item	Complexity	Estimated Effort (man-days)
2.2	Prepare the Project Management Plan	Medium	8
2.3	Set up Jira, GitHub Projects, GitHub repository, and documentation structure	Medium	4
3	Executing		
3.1	Requirement Analysis		22
3.1.1	Analyze business processes and system requirements	Medium	10
3.1.2	Conduct technical and operational feasibility analysis	Medium	6
3.1.3	Research AI-assisted submission verification methods	Complex	6
3.2	System Design		28
3.2.1	Design system architecture and module structure	Complex	8
3.2.2	Design conceptual and logical database models	Complex	8
3.2.3	Design web interfaces and user workflows	Medium	6
3.2.4	Design AI verification and image processing pipeline	Complex	6
3.3	Implementation		176
3.3.1	Authentication and Authorization		14
3.3.1.1	Implement registration, login, logout, and JWT authentication	Medium	6
3.3.1.2	Implement role-based access control	Medium	4
3.3.1.3	Implement user profile management	Medium	4
3.3.2	Contest Management		24
3.3.2.1	Create and configure photography contests	Complex	10
3.3.2.2	Manage contest categories and judging criteria	Medium	6
3.3.2.3	Manage schedules, regulations, awards, and contest status	Complex	8
3.3.3	Participant Submission and Film Metadata Management		28
3.3.3.1	Manage film rolls, frames, cameras, lenses, and film stocks	Complex	10
3.3.3.2	Submit scanned photographs and technical metadata	Complex	10
3.3.3.3	Track registration and submission status	Medium	8

#	WBS Item	Complexity	Estimated Effort (man-days)
3.3.4	Submission Verification and AI Support		34
3.3.4.1	Validate file format, image quality, deadline, and metadata completeness	Medium	6
3.3.4.2	Detect duplicate or highly similar photographs	Complex	10
3.3.4.3	Detect potentially AI-generated images	Complex	10
3.3.4.4	Generate keywords and categorize photographs automatically	Complex	8
3.3.5	Judging and Evaluation Management		26
3.3.5.1	Assign judges by category and judging round	Medium	6
3.3.5.2	Record scores, comments, and evaluation criteria	Complex	10
3.3.5.3	Aggregate scores, manage judging rounds, and calculate rankings	Complex	10
3.3.6	Result Publication and Digital Archive		20
3.3.6.1	Confirm and publish official contest results	Medium	6
3.3.6.2	Preserve award-winning works and technical metadata	Complex	8
3.3.6.3	Manage online exhibitions and historical contest records	Medium	6
3.3.7	Administration and Reporting		18
3.3.7.1	Manage users, roles, permissions, and master data	Complex	8
3.3.7.2	Manage system settings and platform activity logs	Medium	4
3.3.7.3	Generate contest statistics and analytical reports	Complex	6
3.3.8	Cloud Integration and Deployment		12
3.3.8.1	Integrate Firebase Storage for image and digital asset storage	Medium	5
3.3.8.2	Deploy backend services and database infrastructure on Microsoft Azure	Complex	5
3.3.8.3	Set up basic continuous integration and deployment workflow	Medium	2
3.4	Testing		32

#	WBS Item	Complexity	Estimated Effort (man-days)
3.4.1	Unit testing	Complex	8
3.4.2	Integration testing	Complex	8
3.4.3	System testing	Complex	8
3.4.4	User acceptance testing	Complex	8
3.5	Monitoring and Controlling		20
3.5.1	Monitor schedule, scope, backlog, and sprint progress	Complex	10
3.5.2	Track quality, risks, defects, and project performance	Complex	10
4	Closing		18
4.1	Complete technical documentation and final report	Medium	10
4.2	Prepare deployment package and final presentation	Medium	8
Total Estimated Effort (man-days)			330

2.1.2 Project Objectives

The project management objectives are defined as follows:

- Complete the currently assigned documentation from 30 July to 5 August.
- Prepare the assigned report sections in accordance with the structure and level of detail of the reference FPD document.
- Present the current project progress to the supervisor on 10 or 17 August, subject to confirmation.
- Develop the system incrementally so that a functional product version is available at the end of each major iteration.
- Maintain traceability between project requirements, system design, implementation tasks, and testing activities.
- Use Jira and GitHub Projects to monitor backlog items, task status, and sprint progress.
- Use GitHub as the central repository for source code and project documentation.
- Keep critical and high-severity defects at zero before the final release.

The following preliminary defect distribution targets will be used during development and testing.

Table 2.2: Preliminary Defect Distribution Targets

#	Testing Stage	Test Coverage	Maximum Defects	Target Rate	Notes
1	Reviewing	Backend code, database scripts, API contracts, LaTeX documents, and configuration files.	30	5%	Focus on correctness, security, maintainability, and documentation consistency.
2	Unit Testing	Business logic, validation rules, role permissions, scoring calculations, and individual components.	20	3%	Verify each module independently before integration.
3	Integration Testing	Web application, ASP.NET Core Web API, PostgreSQL, Firebase Storage, and AI services.	10	2%	Verify data flow and communication between internal and external services.
4	System Testing	Complete workflows for contest creation, submission, verification, judging, result publication, and archive management.	5	1%	Ensure system consistency, usability, security, and acceptable performance.
5	Acceptance Testing	End-to-end business scenarios performed by representative users.	5	1%	Confirm that the system satisfies the approved business requirements.

2.1.3 Project Risks

Table 2.3: Project Risk Register

#	Risk and Impact	Risk Level	Response Plan
1	Risk: The selected AI-generated image detection model may produce inaccurate results. Impact: Legitimate film photographs may be incorrectly flagged, while manipulated images may remain undetected.	Probability: Medium Severity: High	Use AI results only as decision-support information. Require manual verification by the Contest Organizer before rejecting a submission.

Continued on the next page

Table 2.3 – continued from the previous page

#	Risk and Impact	Risk Level	Response Plan
2	Risk: The image similarity module may produce false-positive duplicate matches. Impact: Different photographs with similar composition or visual content may be incorrectly identified as duplicates.	Probability: Medium Severity: High	Apply similarity thresholds, compare metadata, and allow organizers to review flagged image pairs manually.
3	Risk: AI technologies and models have not yet been officially selected. Impact: The implementation plan and integration effort may change during development.	Probability: High Severity: High	Conduct an early technical evaluation and select models according to accuracy, cost, explainability, and deployment compatibility.
4	Risk: The frontend framework has not yet been confirmed. Impact: User interface implementation and task estimation may be delayed.	Probability: Medium Severity: Medium	Select the frontend framework before the implementation phase and create a small proof of concept before full development.
5	Risk: Large image files may increase storage costs and reduce upload performance. Impact: Participants may experience slow uploads, and cloud service costs may increase.	Probability: Medium Severity: Medium	Define file-size limits, generate compressed previews, store original files securely, and monitor Firebase Storage usage.
6	Risk: Participants may submit incomplete or inaccurate film metadata. Impact: Submission verification and the long-term archival value of photographs may be reduced.	Probability: High Severity: Medium	Define mandatory metadata fields, apply validation rules, and allow organizers to request corrections before judging begins.
7	Risk: Unauthorized access or data leakage may occur. Impact: User information, contest records, and unpublished photographs may be exposed.	Probability: Low Severity: High	Use JWT authentication, role-based access control, secure cloud configuration, audit logs, and encrypted communication.
8	Risk: The project scope may expand because the platform contains many modules. Impact: The team may not finish the core features within the available schedule.	Probability: Medium Severity: High	Prioritize core business flows, maintain a controlled product backlog, and move optional functions to later iterations.
9	Risk: Team members may not complete assigned documentation or development tasks on time. Impact: Dependent report sections or software modules may be delayed.	Probability: Medium Severity: Medium	Track tasks in Jira, review progress weekly, identify blocked tasks early, and redistribute work when necessary.

Continued on the next page

Table 2.3 – continued from the previous page

#	Risk and Impact	Risk Level	Response Plan
10	Risk: Cloud services or external AI services may become temporarily unavailable. Impact: Submission uploads, AI verification, or system deployment may be interrupted.	Probability: Low Severity: Medium	Implement retry mechanisms, error logging, fallback manual review, and regular backups of important data.

2.2 Management Approach

2.2.1 Project Process

The project will apply a combined Scrum and Iterative Development approach. Scrum will be used to organize work into manageable sprint backlogs, while Iterative Development will allow the team to improve each system module through repeated analysis, implementation, review, and testing.

At the beginning of each iteration, the team will select high-priority items from the product backlog. Each item will be divided into smaller tasks and tracked through Jira and GitHub Projects. At the end of an iteration, the team will review the completed work, record unresolved issues, collect feedback, and update the backlog for the next iteration.

The reasons for selecting this approach include:

- Core business functions can be developed and verified before optional functions.
- Requirement changes can be incorporated without redesigning the entire project.
- AI-related features can be researched, tested, and improved in separate iterations.
- Testing can be performed continuously instead of waiting until the end of development.
- Risks can be identified and handled during each sprint.
- The supervisor can review documentation and working increments regularly.
- The team can measure progress through completed backlog items and sprint results.

The preliminary development iterations are organized as follows:

1. **Iteration 1:** Project introduction, management plan, product overview, user requirements, actors, use cases, and functional requirements.

2. **Iteration 2:** Non-functional requirements, requirement appendix, system architecture, and database design.
3. **Iteration 3:** Authentication, user management, and contest configuration.
4. **Iteration 4:** Film roll, frame metadata, participant registration, and submission management.
5. **Iteration 5:** Submission verification and AI-assisted functions.
6. **Iteration 6:** Judging, scoring, ranking, and result publication.
7. **Iteration 7:** Digital archive, reporting, administration, and online exhibition.
8. **Iteration 8:** Integration testing, system testing, deployment, documentation, and final presentation.

2.2.2 Quality Management

Defect Prevention

The team will prevent defects by reviewing requirements before implementation, using clear coding conventions, validating database constraints, and defining acceptance criteria for each backlog item. Security and authorization rules must be considered during design rather than added after implementation.

For AI-assisted functions, test datasets must contain both normal and suspicious submissions. AI output must not be treated as an automatic final decision. Any submission flagged by AI must remain available for manual review.

Reviewing

All important documents, API contracts, database changes, and source code changes must be reviewed before being merged into the main project branch. Review comments must be resolved or formally accepted by the reviewer.

Documentation must be checked for:

- consistency with the approved project proposal;
- correct terminology for film photography and contest management;
- consistency between use cases and functional requirements;
- correct references to actors, modules, and business flows;
- valid LaTeX syntax, table formatting, and chapter structure.

Unit Testing

Unit tests will verify business rules and isolated functions, including:

- authentication and authorization logic;
- contest date and submission deadline validation;
- required film metadata validation;
- scoring and ranking calculations;
- image file validation;
- duplicate similarity threshold processing;
- report calculation and statistical aggregation.

All discovered defects must be recorded in Jira with severity, reproduction steps, expected results, actual results, and the responsible module.

Integration Testing

Integration testing will verify communication among the web application, ASP.NET Core Web API, PostgreSQL, Firebase Storage, Microsoft Azure, and selected AI services. Particular attention will be given to image upload, metadata storage, AI verification requests, judging records, and result publication.

System Testing

System testing will cover complete business flows from contest creation to digital archiving. The system must be tested with multiple simultaneous contests, multiple participants, concurrent judging activities, different user roles, invalid submissions, and interrupted external services.

Acceptance Testing

Acceptance testing will be conducted using representative scenarios for the Administrator, Contest Organizer, Judge, and Participant. A feature will be accepted only when its expected behavior matches the approved requirement and no critical defect remains unresolved.

Table 2.4: Training Plan

Training Area	Participants	When and Duration	Waiver Criteria
Project domain and film photography contest workflow	All team members	Week 1, 2 days	Mandatory
Scrum, Iterative Development, Jira, and GitHub Projects	All team members	Week 1, 2 days	Mandatory
Git and GitHub workflow	All team members	Week 1, 2 days	Mandatory
LaTeX and TeXstudio documentation	All team members	Week 1, 2 days	Mandatory
ASP.NET Core Web API	Assigned backend members	Week 1–2, 7 days	May be waived for members with sufficient practical experience
PostgreSQL database design and administration	Assigned backend and database members	Week 2, 4 days	May be waived after a technical assessment
Firebase Storage and Microsoft Azure deployment	Assigned integration and deployment members	Before cloud integration, 4 days	Mandatory for assigned members
Image similarity detection, AI-generated image detection, and image categorization	Assigned AI members	Before AI implementation, 10 days	Mandatory after the AI technology is selected
Web frontend framework and UI development	Assigned frontend members	To be confirmed	Mandatory after the frontend framework is selected

2.2.3 Training Plan

2.3 Project Deliverables

The current documentation assignment covers the period from 30 July to 5 August. Deliverables after the currently assigned scope are listed according to the reference FPD structure. Their deadlines and responsible members are intentionally left blank until the supervisor and project team confirm the next assignment.

#	Deliverable	Due Date	Assigned Member	Notes
1	Chapter I: Project Introduction	30 July–5 August	Phan Huynh Truc Loan	Prepare the project overview, product background, existing systems, business opportunity, product vision, scope, and limitations.
2	Chapter II: Project Management Plan	30 July–5 August	Nguyen Mai Tuong Vy	Prepare scope estimation, project objectives, risks, management approach, deliverables, responsibility assignments, communication plan, and configuration management.
3	Chapter III: Software Requirement Specification, Part 1	30 July–5 August	Nguyen Tan Phuc	Prepare Product Overview, Context Diagram, User Requirements, and Actors.
4	Chapter III: Software Requirement Specification, Part 2	30 July–5 August	Le Nguyen Quyen Linh	Prepare Use Case Diagram, Use Case Description, and Use Case List.

#	Deliverable	Due Date	Assigned Member	Notes
5	Functional Requirements	30 July–5 August	Nguyen Tan Tai	Prepare detailed functional requirements based on the approved actors, use cases, and business core flows.
6	Non-Functional Requirements			Define security, performance, scalability, availability, usability, storage, and maintainability requirements.
7	Requirement Appendix			Prepare business rules, data definitions, assumptions, constraints, and supporting requirement information.
8	Software Design Description: System Design			Prepare system architecture, module design, component relationships, and external service integration.
9	Software Design Description: Database Design			Prepare conceptual ERD, logical schema, data dictionary, keys, constraints, and relationships.
10	Software Design Description: Detailed Design			Prepare detailed module design, sequence diagrams, API design, and user interface descriptions.
11	Software Testing Documentation			Prepare testing scope, strategy, plan, test cases, defect records, and test reports.

#	Deliverable	Due Date	Assigned Member	Notes
12	Release Package and Installation Guide			Prepare deployment files, environment configuration, database scripts, and installation instructions.
13	User Manual			Prepare user instructions for Administrator, Contest Organizer, Judge, and Participant.
14	Final Capstone Report			Consolidate all approved chapters and supporting project evidence.
15	Supervisor Progress Presentation	10 or 17 August	Project Team	Present the current project progress. The official date is subject to confirmation.
16	Final Presentation and Demonstration			Prepare slides, system demonstration, question-and-answer materials, and final submission package.

2.4 Responsibility Assignments

The responsibility matrix uses the following notation:

- D: Do
- R: Review
- S: Support
- I: Informed

2.5 Project Communications

No regular team meeting will be scheduled on Thursday. The following schedule is preliminary and may be adjusted when members' class schedules are confirmed.

Table 2.6: Responsibility Assignment Matrix

Responsibility	Nguyen Mai Tuong Vy Leader	Phan Huynh Truc Loan	Nguyen Tan Phuc	Le Nguyen Quyen Linh	Nguyen Tan Tai
Project planning and progress tracking	D	S	S	S	S
Prepare Chapter I: Project Introduction	R	D	I	I	I
Prepare Chapter II: Project Management Plan	D	S	I	I	I
Prepare Chapter III, Part 1	R	I	D	S	I
Prepare Chapter III, Part 2	R	I	S	D	I
Prepare Functional Requirements	R	I	S	S	D
Prepare Non-Functional Requirements					
Prepare Requirement Appendix					
Prepare Software Design Description					
Prepare Database Design					
Prepare Detailed Design					
Develop web frontend					
Develop ASP.NET Core Web API					
Develop PostgreSQL database					
Integrate Firebase Storage					
Research and integrate AI modules					
Prepare Testing Documentation					
Deploy the system to Microsoft Azure					
Prepare User Manual					
Prepare Final Report					
Prepare Final Presentation					

2.6 Configuration Management

2.6.1 Document Management

Project documents will be prepared using LaTeX and TeXstudio. GitHub will be used as the central repository for report chapters, diagrams, references, and supporting documentation.

The following document management rules will be applied:

- Each major report chapter will be stored in a separate `.tex` file.
- The main LaTeX file will include the chapter files in the approved report order.
- File names must be descriptive and consistent, for example `Chapter2_Project_Management_Plan.tex`.
- Changes must be committed with clear messages that describe the updated section.
- The project leader will review major structural changes before they are merged.
- Diagrams and images must be stored in dedicated folders and referenced by relative paths.
- Final PDF versions will be generated from the approved main LaTeX branch.

Table 2.7: Project Communication Plan

Communication Item	Who/Target	Purpose	When/Frequency	Tool/Method
Daily progress update	All team members	Report completed work, current tasks, blocked issues, and planned work for the next day.	Daily before 21:30	Messenger or Zalo
Weekly team meeting	All team members	Review backlog, assigned tasks, documentation status, risks, and the following week's plan.	Every Tuesday at 20:00	Google Meet
Documentation review session	Relevant authors and team leader	Check chapter structure, terminology, LaTeX formatting, and consistency among requirements.	Saturday at 19:30 when required	Google Meet and GitHub
Supervisor progress presentation	Supervisor and all team members	Present current documentation and project progress, receive comments, and confirm the next assignment.	10 or 17 August, subject to confirmation	Google Meet or offline meeting
Sprint plan-	All team mem-	Select	At the begin-	Google Meet

2.6.2 Source Code Management

Source code will be managed using Git and GitHub. The repository should use the following branch structure:

- **main**: stable and approved release-ready code;
- **develop**: integrated development version;
- **feature/<feature-name>**: implementation of an individual feature;
- **fix/<issue-name>**: correction of an identified defect;
- **docs/<document-name>**: documentation changes.

Each feature should be developed in a separate branch. Before merging into **develop**, the author must create a pull request and provide a short description of the change. Important changes must be reviewed by at least one other team member.

Jira and GitHub Projects will be used to manage the backlog. Each task should contain a title, description, acceptance criteria, priority, status, and related repository link when applicable.

2.6.3 Tools and Infrastructure

2.6.4 Baseline and Change Control

A baseline will be created after each major document or system increment is reviewed and approved. Any significant change to scope, actors, business flows, database design, or system architecture must be discussed by the team before it is implemented.

A change request should include:

- the requested change;
- the reason for the change;
- affected documents or modules;
- expected impact on schedule and effort;
- decision status;
- related Jira issue and GitHub pull request.

Approved changes will be added to the product backlog and implemented in a future sprint according to priority. Emergency corrections may be handled immediately when they affect data security, authentication, deployment, or critical contest operations.

Table 2.8: Tools and Infrastructure

Category	Tools / Infrastructure
Project management	Jira and GitHub Projects
Source code management	Git and GitHub
Documentation	LaTeX, TeXstudio, and GitHub
Communication	Messenger, Zalo, and Google Meet
Web frontend	To be determined. Jira, GitHub, LaTeX, and TeXstudio are management and documentation tools rather than frontend frameworks.
Mobile application	Not included in the current project scope
Backend	ASP.NET Core Web API
Database	PostgreSQL
Authentication and authorization	JWT authentication with role-based access control
Cloud storage	Firebase Storage
Cloud deployment	Microsoft Azure
AI-generated image detection	Technology or model to be determined
Duplicate and image similarity detection	Technology or model to be determined
Automatic image categorization	Technology or model to be determined
Diagramming	To be determined according to the team's selected design tool
Development environment	Visual Studio, Visual Studio Code, TeXstudio, and local PostgreSQL tools as required

3

Software Requirement Specification

3.1 Product Overview

The AI-powered Film Photography Contest Management Platform is a web and mobile application designed to support the organization and management of film photography contests. The platform provides a centralized environment where administrators, contest organizers, judges, and participants can perform contest-related activities efficiently.

The system supports the entire contest lifecycle, including contest creation, participant registration, photograph submission, judging management, result publication, and digital archiving. Participants can upload scanned film photographs together with technical information such as film stock, camera, lens, ISO, film format, frame number, shooting location, developing laboratory, and scanning specifications.

To improve contest management, the platform integrates Artificial Intelligence (AI) for submission verification and analysis. AI technologies help detect duplicate or highly similar photographs, identify AI-generated images, automatically categorize photographs, generate image keywords, and produce statistical reports for contest organizers.

By integrating contest management, judging workflows, AI-assisted verification, and digital asset management into a single platform, the system improves management efficiency, enhances transparency throughout the judging process, and preserves valuable film photography records for future contests, exhibitions, and educational activities.

3.2 Context Diagram

The context diagram illustrates the interactions between the AI-powered Film Photography Contest Management Platform and its external entities,

including the Administrator, Contest Organizer, Participant, Judge, and AI Services.

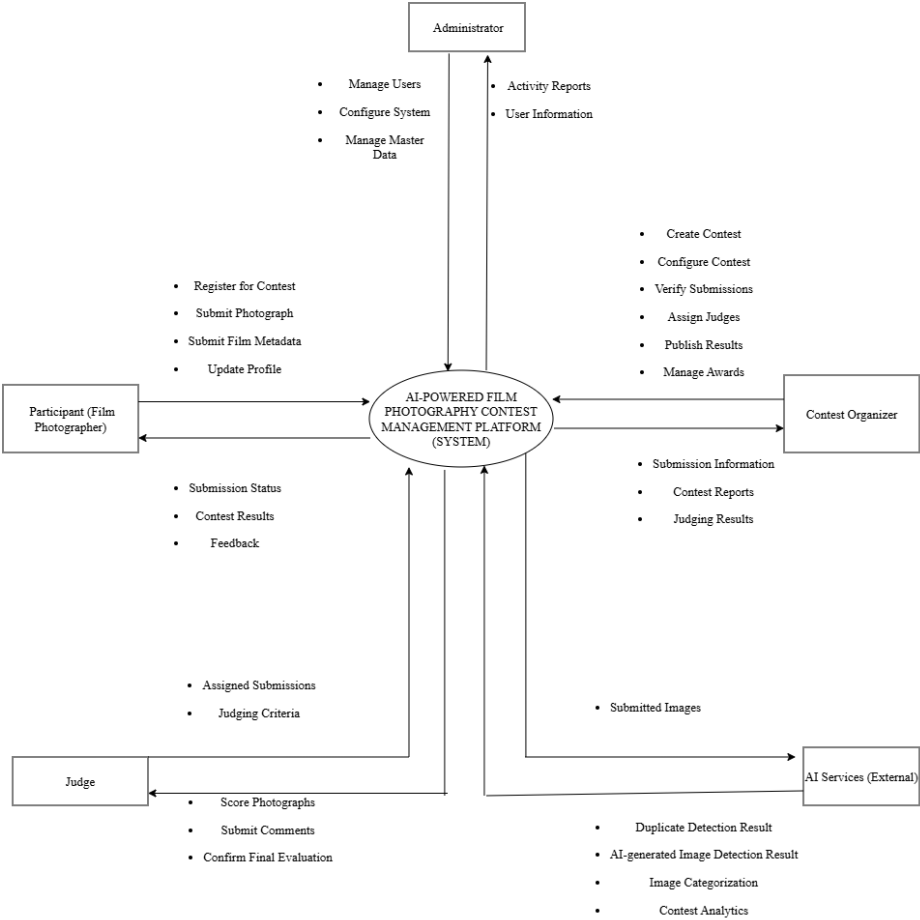


Figure 3.1: Context Diagram of the AI-powered Film Photography Contest Management Platform

3.3 User Requirements

This section describes the requirements of the primary users of the AI-powered Film Photography Contest Management Platform. The system shall provide appropriate functionalities to support contest management, photograph submission, judging, and AI-assisted verification.

- Participants shall be able to create an account, log in securely, and manage their personal profiles.

- Participants shall be able to register for photography contests and submit photographs together with the required metadata before the submission deadline.
- Participants shall be able to monitor the status of their submissions, receive feedback, and view contest results after publication.
- Judges shall be able to access assigned submissions, evaluate photographs according to the judging criteria, and submit scores and comments.
- Contest Organizers shall be able to create contests, configure contest rules, verify submissions, assign judges, and publish contest results.
- Administrators shall be able to manage user accounts, configure system settings, and monitor overall system activities.
- The system shall integrate AI services to detect duplicate photographs, identify AI-generated images, and categorize submitted photographs for contest management.

3.4 Actors

The AI-powered Film Photography Contest Management Platform involves several actors who interact with the system to perform different tasks.

- **Administrator**
 - Manages user accounts.
 - Creates and manages contests.
 - Reviews submissions and system reports.
 - Configures contest rules and schedules.
- **Contest Organizer**
 - Creates photography contests.
 - Defines contest requirements.
 - Reviews participant submissions.
 - Publishes contest results.
- **Participant**
 - Registers an account.
 - Joins photography contests.
 - Uploads photographs.

- Views contest results and feedback.

- **Judge**

- Reviews submitted photographs.
- Scores participant submissions.
- Provides comments and feedback.

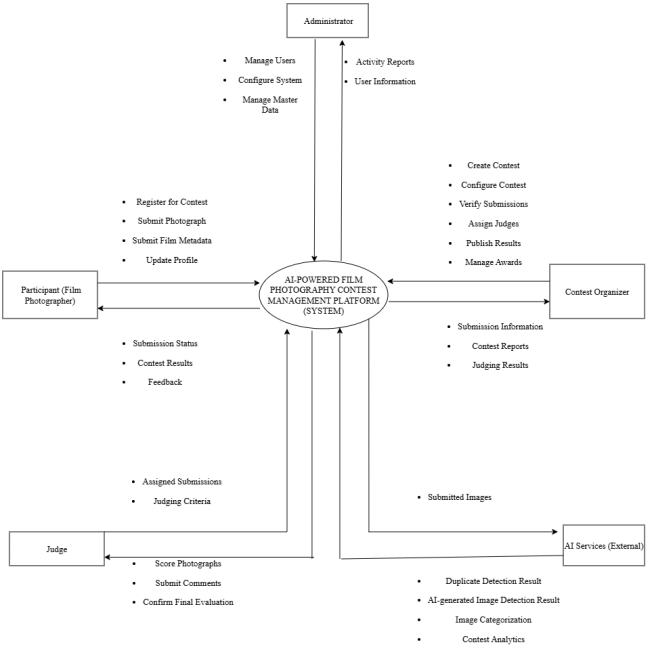


Figure 3.2:

“This is a quote and I don’t know who said this.”

– Author’s name, *Source of this quote*

3.5 Section heading

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem ipsum list:

1. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
2. Duis ac mi magna, a consectetur elit.
3. Curabitur posuere erat *dignissim ligula euismod* ut euismod nisi.
4. Fusce vulputate facilisis neque, et ornare mauris mattis vel.
5. Mauris sit amet nulla mi, vitae rutrum ante.
6. Maecenas quis nulla risus, vel tincidunt ligula.
7. Nullam ac enim neque, non *dapibus* mauris.
8. Integer volutpat leo a orci suscipit eget rhoncus urna eleifend.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna¹:

3.5.1 Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam. Etiam ullamcorper, nunc a accumsan adipiscing, turpis odio bibendum erat, id convallis magna eros nec metus. Sed vel ligula justo, sit amet vestibulum dolor. Sed vitae augue sit amet magna ullamcorper suscipit. Quisque dictum ipsum a sapien egestas facilisis.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus

¹ Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus.

rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna.

3.5.2 Lorem ipsum dolor sit amet, consectetur adipiscing.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam.

3.5.3 Lorem ipsum dolor sit amet

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet² erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam. Etiam ullamcorper, nunc a accumsan adipiscing, turpis odio bibendum erat, id convallis magna eros nec metus. Sed vel ligula justo, sit amet vestibulum dolor. Sed vitae augue sit amet magna ullamcorper suscipit. Quisque dictum ipsum a sapien egestas facilisis.

In hac habitasse platea dictumst. Nullam turpis erat, porttitor ut pretium ac, condimentum sed dui. Praesent arcu elit, tristique sit amet viverra at, auctor quis tortor. Etiam eleifend posuere aliquam. Donec sed mattis sapien. Aenean urna arcu, suscipit at rutrum ac, adipiscing ac felis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Praesent condimentum felis a ipsum ullamcorper semper. Vivamus eu odio sem, dictum luctus nunc. Etiam tincidunt venenatis dolor non pellentesque.

3.5.4 Lorem ipsum dolor sit amet, auctor et pulvinar non

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam. Etiam ullamcorper, nunc a accumsan adipiscing, turpis odio bibendum erat, id convallis magna eros nec metus. Sed vel ligula justo, sit amet vestibulum dolor. Sed vitae augue sit

²www.example.com

amet magna ullamcorper suscipit. Quisque dictum ipsum a sapien egestas facilisis.

In hac habitasse platea dictumst. Nullam turpis erat, porttitor ut pretium ac, condimentum sed dui. Praesent arcu elit, tristique sit amet viverra at, auctor quis tortor. Etiam eleifend posuere aliquam. Donec sed mattis sapien. Aenean urna arcu, suscipit at rutrum ac, adipiscing ac felis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Praesent condimentum felis a ipsum ullamcorper semper. Vivamus eu odio sem, dictum luctus nunc. Etiam tincidunt venenatis dolor non pellentesque.

3.6 Another section heading

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Table 3.1: Sample table

S. No.	Column#1	Column#2	Column#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem ipsum list:

- Mauris sit amet nulla mi, vitae rutrum ante.
- Maecenas quis nulla risus, vel tincidunt ligula.
- Nullam ac enim neque, non *dapibus* mauris.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna³:

³Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus.

3.6.1 Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam. Etiam ullamcorper, nunc a accumsan adipiscing, turpis odio bibendum erat, id convallis magna eros nec metus. Sed vel ligula justo, sit amet vestibulum dolor. Sed vitae augue sit amet magna ullamcorper suscipit. Quisque dictum ipsum a sapien egestas facilisis.

3.6.2 Lorem ipsum dolor sit amet, consectetur adipiscing

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis risus ante, auctor et pulvinar non, posuere ac lacus. Praesent egestas nisi id metus rhoncus ac lobortis sem hendrerit. Etiam et sapien eget lectus interdum posuere sit amet ac urna. Aliquam pellentesque imperdiet erat, eget consectetur felis malesuada quis. Pellentesque sollicitudin, odio sed dapibus eleifend, magna sem luctus turpis, id aliquam felis dolor eu diam.